

By **Jayant Sinha**

THE DANGERS OF **TECHNO-COLONIALISM**

INDIA NEEDS HUGE INVESTMENTS IN FRONTIER TECHNOLOGIES TO BE GLOBALLY COMPETITIVE. BUILDING OUR OWN DIGITAL INFRASTRUCTURE IS KEY OR WE RISK DEPENDENCE ON GLOBAL POWERS

Building a Viksit Bharat requires building a Harit Bharat as well. To deliver sustainable prosperity for the Indian people, we must not only position our economy at the global competitiveness frontier but also restore our environment. We can achieve this green frontier development model only through accelerated technological progress powered by massive financial investments.

The green frontier model is not defined by labour arbitrage or gleaming infrastructure. Instead, it is defined by the ability to deploy frontier technologies at speed and scale. It is defined by the ability to drive resource efficiency across the economy. The United States and China are rapidly shaping the technology standards, global value chains, energy grids and capital flows that will define the next era of global competition. If we do not keep up, we run the risk of being fully dependent on foreign technology, equipment and capital.

The capital requirements to stay in the game are enormous. In AI alone, the United States is committing $500 billion (Rs 45 lakh crore) annually across public and private channels to artificial intelligence, advanced computing infrastructure, data centres, clean power and associated supply chains. It is also investing hundreds of billions of dollars in space technology, semiconductors, biosciences and other advanced technologies. China is investing $100 billion (Rs 9 lakh crore) or more annually in AI and adjacent frontier technologies, alongside sustained investment in clean energy, batteries and advanced manufacturing. Both countries see winning the technology investment race as essential for long-term economic and strategic leadership.

India can't reach the global competitiveness frontier without massively hiking investments in frontier technologies. AI is only one part of the picture. India must also invest in quantum computing, biosciences, space technologies, advanced materials and both fission and fusion energy. Each needs long-duration capital and deep tolerance for technological risk.

At the same time, India's net zero pathway demands sustained investment across power, mobility, industry and urban systems. Renewable capacity must continue to expand. Transmission, distribution and storage infrastructure must scale rapidly. Hard-to-abate sectors such as steel, cement and chemicals remain capital-intensive and technologically immature, and require cleaner production pathways. These investments generate returns over long horizons and cannot be financed through short-term instruments.

Taken together, the scale of the challenge becomes clear. India's corporate capital expenditure has remained relatively steady in recent years at about $100 billion per year, even as balance sheets have strengthened and macroeconomic stability has improved. To reach the green frontier, corporate investment alone must rise by at least $100–200 billion (Rs 9-18 lakh crore) per year on an incremental basis. Note that we need to invest tens of billions of dollars annually in AI infrastruc-

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